Value added problem 2
Name-Archit Shelke
Prn-25070521227
Section-D

Problem Statement:
Given a string s, composed of different combinations of '(' , ')', '{', '}', '[', ']'. Determine whether the Expression is balanced or not. An expression is balanced if:
1. Each opening bracket has a corresponding closing bracket of the same type.
2. Opening brackets must be closed in the correct order.

Concept: Stack
Constraints: 1 ≤ s.size() ≤ 106
s[i] ∈ {'{', '}', '(', ')', '[', ']'}

Code:
```
#include <stdio.h>
#include <string.h>
#define MAX 1000006
char stack[MAX];
int  top = -1;
void push(char c)  { stack[++top] = c; }
char pop()         { return top == -1 ? '\0' : stack[top--]; }
int  isEmpty()     { return top == -1; }
int isMatch(char open, char close) {
   return (open == '(' && close == ')') ||
          (open == '{' && close == '}') ||
          (open == '[' && close == ']');
}
int isBalanced(char* s) {
   top = -1;
   for (int i = 0; s[i]; i++) {
      if (s[i] == '(' || s[i] == '{' || s[i] == '[')
         push(s[i]);
      else {
         if (isEmpty() || !isMatch(pop(), s[i]))
            return 0;
      }
   }
   return isEmpty();
}
int main() {
   char s[MAX];
   printf("Name - Archit Shelke\n");
   printf("PRN - 25070521227\n");
```

```
    printf("Section - D\n");
    printf("Batch - 2025-29\n\n");
    printf("Enter bracket string: ");
    scanf("%s", s);
    printf("Output: %s\n", isBalanced(s) ? "true" : "false");
    return 0;
}
```

Output:

Output   Clear

```
Name - Archit Shelke
PRN - 25070521227
Section - D
Batch - 2025-29

Enter bracket string: [{()}]
Output: true


=== Code Execution Successful ===
```